

PhishGuard - Research Review

Research review | September 2026

The question is whether simple text features distinguish phishing-like language from legitimate messages under a distribution change. This is AI-assisted, synthetic educational work, not a deployed detector.

Why the first 100% result was not valid evidence of generalisation

The original random split put variants of the same small set of templates on both sides. Names and invoice numbers changed, but wording strongly predicted the label. The score could correctly describe that split while being invalid as a real-world performance claim. This is template leakage/dependence and a dataset shortcut, not proof that the code fitted the vectoriser on test labels. The current pipeline fits TF-IDF inside each training fold.

Current design

research.py groups the 240 training messages by their 16 body templates and uses four stratified group folds. Both word (1-2 gram) and character-within-word (3-5 gram) pipelines use the same splits and a 3,000-feature cap. Each fold fits its own TF-IDF vocabulary and logistic regression. Fold predictions are saved.

Thresholds 0.10 to 0.90, step 0.05, are compared using training out-of-fold F1. Ties prefer the threshold nearest 0.5. Representation selection uses mean training-fold F1 at 0.5, with word features winning ties. Neither selection reads challenge labels. These are selection scores, not nested-CV estimates.

Then each representation is fitted on all training messages and evaluated on 20 fixed synthetic challenge messages. Saved ROC/PR coordinates, AUC, average precision and message-level errors make the comparison inspectable. The challenge was previously inspected and is not a new blind external test.

Actual results and interpretation

Both representations obtain F1=1.00 in every training fold. Grouping bodies does not remove recurring subject templates or all language shortcuts. Do not advertise the CV score as realistic accuracy.

On the challenge, word features: accuracy .65, F1 .6316, ROC AUC .70; character features: accuracy .45, F1 .2667, ROC AUC .41. Both selected threshold 0.5. Character features are not automatically better: they fragment words into patterns and can still learn synthetic shortcuts. This comparison does not establish superiority across corpora or hyperparameters.

Error review

results/research/predictions.csv contains full synthetic text, scores, labels and false-positive/false-negative categories for both models. Inspect messages side by side; coefficients describe association, not causal explanations. Legitimate security notices share phishing vocabulary, while indirect requests may lack urgent/password wording. These are hypotheses to check against each message, not verified causes of every error. Labels themselves can be ambiguous without sender identity, links, headers or conversation context.

Remaining risks

- No external, time-separated or campaign-disjoint test has been run.
- Only 20 balanced challenge messages: unstable estimates and unrealistic prevalence.
- Body groups still share subject templates. Fully grouping both can connect most examples into a few groups; a newly collected dataset is preferable.

- Threshold maximises F1, not a specified cost of false alarms/missed attacks.
- Scores are uncalibrated; a value of .8 is not an established 80% real-world risk.
- Only text is modelled; URLs, attachments, sender reputation and transport are absent.
- Duplicate normalised train/test text and shared explicit groups are rejected; semantic near-duplicates and time leakage still need dataset-specific checks.
- Dependency ranges permit version drift; run records include local versions.

External data contract

Supply licensed, de-identified CSV files with message_id, subject, body, binary label (0 legitimate, 1 phishing), and training group (campaign/sender). Supply group in both files to check group overlap. --schema mapping.json accepts column_map (source name to canonical name) and label_map (source labels to 0/1), applied to both files. Mixed schemas should be normalised first. Do not use row IDs as groups. Record source, licence, collection dates, label procedure and group rationale before comparing results. No corpus is downloaded or uploaded by the loader.

References: scikit-learn cross-validation (https://scikit-learn.org/stable/modules/cross_validation.html) and data leakage guidance (https://scikit-learn.org/stable/common_pitfalls.html).

Code walkthrough

Read INTERVIEW_WALKTHROUGH.md in the repository for the file map, design explanations, likely interview questions and practical exercises.